Do Register Tokens Regularize Vision Transformers Under Data Scarcity?
A Controlled 24-Run Empirical Ablation on Low-Data CIFAR-100

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Team Task Distribution & Course §6 Ownership

Shahin Alakparov

Core Architecture & Training Lead

`src/models/register_vit.py`, `scripts/train.py`
Core wrapper, positional invariant, PyTorch AMP training.

Gulnisa Abdurahmanli

Data Engineering & Multi-Budget Lead

`src/data/cifar100_subset.py` (100 & 300 imgs/class)
Stratified sampling, deterministic splits, seed isolation.

Narmina Ibrahimova

Metrics & Interpretability Lead

`src/models/attention_hook.py`, `src/metrics/`
Unfused hooks, Shannon entropy $\bar{H}^{(l)}$, outlier rate $\rho^{(l)}$.

Emil Ahmedli

Ablation Sweeps & Hardware Lead

`scripts/run_*sweep.sh`, `configs/`, `team1.conf`
NVIDIA A100 GPU cluster, WireGuard VPN, 24 sweep runs.

Rufet Dosteliyev

Ablation Analytics, Figures & Paper Lead

`src/utils/aggregate_databudget.py`, `plot_*.py`, `paper/`
Statistical reduction, publication figures, IEEE paper writing.

Equal Contribution Policy (§6 Compliance)

Strict parity across formulation, cluster compute, diagnostics, statistical analytics, and paper deliverables.

Large ViTs quietly vandalise their own attention.

- Darcet et al. (ICLR 2024): in foundation vision transformers, a handful of background patch tokens acquire enormous activation norms. The `[CLS]` token routes much of its attention to them.
- Those tokens sit in empty image regions and carry global context — the model repurposes the image as scratch paper.
- Their fix is architectural and almost free: give the model dedicated scratch tokens (*registers*), and the artifacts disappear.

The gap in that story

Darcet demonstrated this where data is abundant. If registers absorb capacity the model would otherwise spend memorising, they should behave like a **structural regularizer** — and a regularizer only proves itself where overfitting bites.

Our core research question

When a ViT is starved of data, do registers reduce the generalization gap — and does that transfer to held-out test performance?

Three Hypotheses, Each Falsifiable

H1: Regularization

Registers shrink the generalization gap:

$$\Delta\mathcal{L} = \mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}}$$

and lower validation loss by acting as an architectural bottleneck.

H2: Capacity Dilution

At $d = 192$ (ViT-Tiny), a large K eats representational margin across softmax rows. Accuracy becomes non-monotonic in K.

H3: Entropy Collapse

Registers prevent the layer-wise collapse of attention entropy $\bar{H}^{(l)}$ and suppress high-norm patch outliers.

Why this is worth an experiment: Nobody has tested registers as a regularizer, at compact scale, under strict data budgets. A clean answer is valuable whichever way it falls.

Two controlled low-data regimes on CIFAR-100:

- **100 images/class (Primary):** 9,000 train / 1,000 val / 10,000 test. Strict data scarcity.
- **300 images/class (Extension):** 27,000 train / 3,000 val / 10,000 test. Tests scale-invariance.
- Upsampled to 224×224 so the pretrained patch grid ($14 \times 14 = 196$ patches) applies unchanged.
- RandomResizedCrop, horizontal flip, AutoAugment; evaluation uses deterministic resize and centre crop.

The design that makes it a controlled test

The subset is drawn under the run's own seed. So a seed is a replicate of the **whole pipeline**, not just weight initialization.

Within-seed pairing

Within each seed, all four register arms ($K \in \{0, 1, 4, 8\}$) see the **identical images** in the **identical batch order**.

One Thing Moves, and Only One

Held fixed across every single arm:

- The exact same training images
- The exact same train/validation split
- The exact same batch order
- The exact same 50-epoch schedule, AdamW (5×10^{-4}), cosine decay, AMP, and batch size 64

Varied: $K \in \{0, 1, 4, 8\}$, the number of register tokens. **Nothing else.**

Why pairing matters at $n = 3$

Between-seed spread in validation accuracy is **2.17 pp —** far larger than any effect registers could plausibly have.

Because the arms are strictly matched within each seed, we can **subtract that noise away** via paired differences ($\bar{\delta} = X_K - X_0$) instead of drowning in it.

The token sequence, and the slice that decides everything:

`[CLS]` $\parallel$ $r_1, r_2, \ldots, r_K$ (K registers, no pos. embed) $\parallel$ $p_1, p_2, \ldots, p_{196}$ (196 spatial patches)

Architectural Sequencing:

- Registers join every self-attention block and are discarded before the classification head.
- Spatial patches start strictly at index $1 + K$.
- Parameter cost at $K = 8$: 1,536 of 5.54 M parameters — a relative increase of just 2.8×10^{-4}.

The off-by-K trap

Slice the spatial row from index 1 instead of $1 + K$ and register columns leak into the attention map. It still looks plausible — which is exactly why our test suite rigorously asserts slice width across all depths.

Two Diagnostic Mechanism Readouts

Layer-wise Attention Entropy

$$\bar{H}^{(l)} = -\frac{1}{BHS} \sum_{b,h,i,j} A^{(l)}_{b,h,i,j} \log_2 A^{(l)}_{b,h,i,j}$$

- Attention rows are probability distributions; entropy measures routing concentration.
- Artifacts appear as entropy collapse in deep blocks.
- Modern `timm` executes fused attention; our hook reconstructs the exact $[B, H, S, S]$ softmax matrix from unfused Q, K.

Patch-Norm Outlier Rate

$$\rho^{(l)} = \frac{1}{BN} \sum_{b,n} \mathbf{1}\left[\|x^{(l)}_{b,1+K+n}\|_2 > \mu^{(l)}_b + 3\sigma^{(l)}_b\right]$$

- The exact 3σ artifact definition: fraction of spatial patch tokens with anomalously large L_2 norm.
- Evaluated on the residual stream, preventing LayerNorm squashing.

Both metrics are read from the identical 64 test images across all runs for perfect input parity.

04. Sweep Execution — Emil (Ablation Sweeps & Hardware Lead)

The 24-run controlled study matrix:

- **100 pc baseline:** $K \in \{0, 1, 4, 8\} \times$ seeds $\{42, 1337, 3407\}$ (12 runs).
- **300 pc scaling:** $K \in \{0, 1, 4, 8\} \times$ seeds $\{42, 1337, 3407\}$ (12 runs).
- Executed on remote NVIDIA A100-SXM4 GPU cluster (20GB MIG partition) via WireGuard VPN.
- **100% execution success:** 24/24 runs completed cleanly; zero OOMs, zero crashes.

Isolation & Artifact Completeness

Each run writes its own isolated directory:

- `metrics.json`, `best_model.pth`
- `last_model.pth`, `train_history.csv`

Three seeds can never overwrite each other's weights.

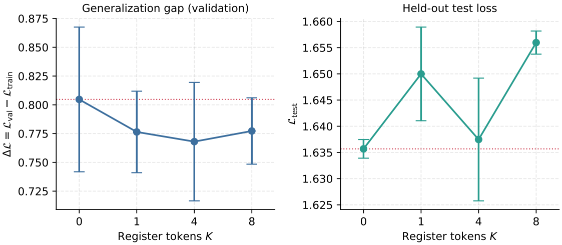
Guards against manual error

The table generator independently recomputes all statistics from raw logs. Not one number was typed by hand.

Configuration	Test Top-1 (%)	Test Loss	Val Loss	Gap $\Delta\mathcal{L}$
Baseline ($K = 0$)	$\mathbf{75.19 \pm 0.24}$	$\mathbf{1.6357 \pm 0.0018}$	1.6829 ± 0.0609	0.8047 ± 0.0628
Registers ($K = 1$)	74.66 ± 0.42	1.6500 ± 0.0089	1.6545 ± 0.0380	0.7764 ± 0.0354
Registers ($K = 4$)	75.06 ± 0.33	1.6375 ± 0.0117	$\mathbf{1.6492 \pm 0.0536}$	$\mathbf{0.7680 \pm 0.0514}$
Registers ($K = 8$)	74.67 ± 0.12	1.6560 ± 0.0022	1.6602 ± 0.0268	0.7773 ± 0.0288

The winners split down the middle:

- The control baseline takes both held-out test columns (75.19%, loss 1.6357).
- $K = 4$ takes both validation-derived columns (1.6492, gap 0.7680).
- Paired per seed: $K = 1$ and $K = 8$ lose accuracy in **3 of 3** seeds. $K = 4$ wins in only 1 of 3 ($\bar{\delta} = -0.13$ pp, $p = 0.466$).
- Registers cost between zero and half a point of accuracy; they gain nothing on held-out data.



$K = 4$ vs. control, paired over seeds:

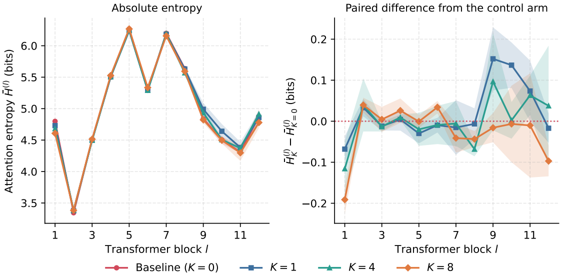
	$\bar{\delta}$	Dir.	p
Val loss	-0.0337	3/3	**0.024**
Gen. gap	-0.0367	3/3	**0.051**
Test loss	$+0.0018$	2/3	**0.823**

Why it dies: Final training loss across all arms is 0.8782, 0.8780, 0.8812, 0.8829 — a spread of just 0.005.

Nothing was traded

A true regularizer trades training fit for held-out fit. Training loss never moved.

Mechanism Checks: H2 Supported, H3 Not Supported

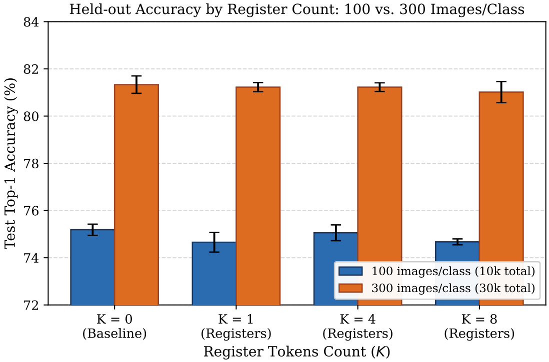


H2 — Capacity Dilution Supported:

- $K = 8$ produces the study's only consistent held-out effect: test loss $+0.0203$, **3 of 3 seeds**, $p = 0.014$.
- Final-block attention entropy also falls in 3 of 3.

H3 — Entropy Collapse Not Supported:

- Entropy curves lie directly on top of each other ($\Delta H \approx 0.001$ bits).
- Outlier rate goes 1.227% $\to$ 1.254% as K grows — it rises slightly, opposite of artifact absorption.



Is the null result an artifact of extreme starvation?

- Tripling data to 300 pc (27k train) surges baseline accuracy by **+6.14 pp** (75.19% → 81.33%).
- Yet registers still provide **zero benefit**:
  - $K = 0$ (Control): $\mathbf{81.33 \pm 0.37\%}$
  - $K = 1$: $81.23 \pm 0.19\%$
  - $K = 4$: $81.23 \pm 0.18\%$
  - $K = 8$: $81.02 \pm 0.45\%$
- All pairwise differences are non-significant ($p > 0.50$).
- **Conclusion:** The null result is **scale-invariant** across low-data budgets.

07. Bonus Experiment: Registers Without Training

Do registers even need to be trained? Jiang et al. (NeurIPS 2025) redirect outlier neurons into an untrained slot at inference time. We ran an exploratory single-block test on our $K = 0$ checkpoints:

Condition	Test Top-1 (%)	Loss	Mechanism / Insight
Baseline ($K = 0$, untouched)	75.18 ± 0.30	1.1110	Control baseline model
Stage 1: Passive empty slot	75.25 ± 0.24	1.1103	$+0.07$ pp (inert slot is harmless)
Stage 2: Redirection (Block 1)	75.25 ± 0.26	1.1104	Top-4 neurons redirected into slot
Stage 2: Redirection (Block 9)	75.24 ± 0.25	1.1103	Top-4 neurons redirected into slot

Architectural note on Block 11: Block 11 is followed by no further self-attention; the classification head reads solely `[CLS]`. Any intervention on non-CLS tokens at Block 11 is architecturally incapable of affecting predictions. Across valid blocks, redirection does not exceed the passive slot.

Interpretation: A Scope Result, Not a Refutation

Regime Mismatch:

- In Darcet et al., registers were present throughout pretraining at foundation scale.
- Here, registers are added to an ImageNet-pretrained backbone and given 50 epochs over 9k–27k images.
- The routing never reorganised — entropy profiles remain the pretrained profile.

Artifacts Present but Unmoved:

- Control outlier rate (1.23%) is $9\times$ the Gaussian 3σ rate: heavy-tailed structure is real.
- But registers simply do not absorb it here.
- **And $K = 8$ is not free:** 8 content-free keys in a 205-token sequence at $d = 192$ costs held-out accuracy and loss.

Conclusion & Defense Takeaways

1. **No held-out benefit:** Across 24 controlled runs, no register configuration outperforms the register-free control on held-out test data.
2. **Validation illusion:** The apparent regularization signal on validation ($p = 0.024$) is an artifact of checkpoint selection on a noisy split; training loss was unmoved.
3. **Scale-invariance confirmed:** Tripling data to 300 pc ($+6.14$ pp gain) preserves the exact null result; test-time registers also produce no gain.
4. **Methodological lesson:** Never infer generalization from the split that performs model selection.

Full reproducibility: Code, sweep runners, and generators available at:
`https://github.com/shahin1717/vit-research`

Thank You!

Questions & Oral Defense Discussion

Artifacts & Codebase: 72/72 tests passing • Pinned dependencies • 24 A100 runs completed